

(SOLUTIONS) EXAM IV

CALCULUS I, FALL 2017

Problem 1

Part (a)

In general, the equation of the line that represents the best linear approximation to $f(x) = x^4 + 3x^2$ at the point $x = a$ is given by:

$$\begin{aligned}y &= f'(a)(x - a) + f(a) \\y &= (4a^3 + (3)(2)a)(x - a) + (a^4 + 3a^2) \\y &= (4a^3 + 6a)(x - a) + (a^4 + 3a^2).\end{aligned}$$

So, in our case where $a = -1$, we have

$$\begin{aligned}y &= f'(-1)(x - (-1)) + f(-1) \\y &= (4(-1)^3 + 6(-1))(x + 1) + ((-1)^4 + 3(-1)^2) \\y &= -10x - 6.\end{aligned}$$

Part (b)

Using our linear approximation, we have

$$f(-0.9) \approx -10(-0.9) - 6 = 3.$$

In fact, direct computation shows that $f(-0.9) = 3.0861$, so our error is only 0.0861.

Problem 2

Part (a)

As limit given produces the indeterminate form $\frac{0}{0}$, we apply L'Hôpital's rule as follows:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{3(x+1) - 3\cos(x)}{x} &= \lim_{x \rightarrow 0} \frac{3 + 3\sin(x)}{1} \\&= 3 + 3\sin(0) = 3\end{aligned}$$

Part (b)

As the function inside of this limit is a rational function, and as the degree of the polynomial function in the numerator is equal to the degree of the polynomial function in the denominator, the rational function approaches the ratio of the leading coefficients of these polynomials as x approaches infinity. That is,

$$\lim_{x \rightarrow \infty} \frac{4 + 6x - 2x^4}{3x^4 + 5x^3 - 7x^2 + 4x + 27} = -\frac{2}{3}.$$

Part (c)

As limit given produces the indeterminate form $\frac{0}{0}$, we apply L'Hôspital's rule as follows:

$$\lim_{x \rightarrow 1} \frac{x^n - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{nx^{n-1}}{1} = n$$

Part (d)

First, we manipulate the function inside the limit algebraically:

$$\lim_{x \rightarrow 0^+} x \ln(x) = \lim_{x \rightarrow 0^+} \frac{\ln(x)}{\left(\frac{1}{x}\right)}$$

Now, we have a limit which produces the indeterminate form $\frac{-\infty}{\infty}$. As such, we apply L'Hôspital's rule as follows:

$$\begin{aligned} \lim_{x \rightarrow 0^+} x \ln(x) &= \lim_{x \rightarrow 0^+} \frac{\ln(x)}{\left(\frac{1}{x}\right)} \\ &= \lim_{x \rightarrow 0^+} \frac{\left(\frac{1}{x}\right)}{\left(-\frac{1}{x^2}\right)} \\ &= \lim_{x \rightarrow 0^+} \left(\frac{1}{x}\right)(-x^2) \\ &= \lim_{x \rightarrow 0^+} -x \\ &= 0 \end{aligned}$$

Problem 3**Part (a)**

The possible correct choices are (i.) and (iv.)

Part (b)

If you chose (i.), the calculation is as follows:

$$\begin{aligned} f'(3) &= \lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3} \\ &= \lim_{x \rightarrow 3} \frac{2x^2 - 2(3)^2}{x - 3} \\ &= \lim_{x \rightarrow 3} \frac{2x^2 - 18}{x - 3} \\ &= \lim_{x \rightarrow 3} \frac{2(x^2 - 9)}{x - 3} \\ &= \lim_{x \rightarrow 3} \frac{2(x - 3)(x + 3)}{x - 3} \\ &= \lim_{x \rightarrow 3} 2(x + 3) \\ &= 2(3 + 3) = 12 \end{aligned}$$

If you chose (*iv.*), the calculation is as follows:

$$\begin{aligned}f'(3) &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} \\&= \lim_{x \rightarrow 3} \frac{2(3+h)^2 - 2(3)^2}{h} \\&= \lim_{x \rightarrow 3} \frac{2(3^2 + 6h + h^2) - 18}{h} \\&= \lim_{x \rightarrow 3} \frac{18 + 12h + 2h^2 - 18}{h} \\&= \lim_{x \rightarrow 3} \frac{12h + 2h^2}{h} \\&= \lim_{x \rightarrow 3} 12 + 2h \\&= 12 + 2(0) = 12\end{aligned}$$

Problem 4

To find the absolute maximum and minimum values of $A(x)$ on the interval $[0, 50]$, let us first note that $A(x)$ is continuous. As such, the extreme value theorem tells us that $A(x)$ achieves both its absolute maximum and absolute minimum on $[0, 50]$, and must do so either at a critical point or an endpoint of the interval. Moreover, as $A(x)$ is a polynomial, its derivative is defined everywhere, so the only critical points of $A(x)$ will occur when $A'(x) = 0$. Calculating, $A'(x) = 100 - 4x$, so

$$\begin{aligned}A'(x) &= 0 \\100 - 4x &= 0 \\4x &= 100 \\x &= 25.\end{aligned}$$

So, the only critical point of $A(x)$ is $x = 25$, which is in the interval $[0, 50]$. With this information in hand, we test our endpoints and our critical point:

$$A(0) = (100 - 2(0))(0) = 0, \quad A(50) = (100 - 2(50))(50) = 0, \quad A(25) = (100 - 2(25))25 = 1250.$$

Thus, the absolute maximum of $A(x)$ on the interval $[0, 50]$ is 1250 and occurs when $x = 25$. Similarly, the absolute minimum of $A(x)$ on the interval $[0, 50]$ is 0 and occurs when either $x = 0$ or $x = 50$.

Problem 5

To find the average value of the function $f(x) = x(x - 1)$ on the interval $[0, 2]$, we use the definition of the average value and compute directly as follows:

$$\begin{aligned}\bar{f} &= \frac{1}{2-0} \int_0^2 f(x) dx \\ &= \frac{1}{2} \int_0^2 x(x-1) dx \\ &= \frac{1}{2} \int_0^2 x^2 - x dx \\ &= \frac{1}{2} \left(\frac{1}{3} x^3 - \frac{1}{2} x^2 \right) \Big|_0^2 \\ &= \frac{1}{2} \left[\left(\frac{1}{3} 2^3 - \frac{1}{2} 2^2 \right) - \left(\frac{1}{3} 0^3 - \frac{1}{2} 0^2 \right) \right] \\ &= \frac{1}{2} \left(\frac{2}{3} \right) \\ &= \frac{1}{3}\end{aligned}$$

Problem 6

Skip this one.

Problem 7**Part (a)**

To find the velocity of the particle at time t , we make use of the Fundamental Theorem of Calculus as follows:

$$\begin{aligned}v(t) &= v(0) + \int_0^t a(x) dx \\ &= 2 + \int_0^t 1 - \cos(x) dx \\ &= 2 + (x - \sin(x)) \Big|_0^t \\ &= 2 + [(t - \sin(t)) - (0 - \sin(0))] \\ &= 2 + t - \sin(t)\end{aligned}$$

So, the velocity of the particle at time t is $(2 + t - \sin(t))$ ft/s.

Part (b)

To find the position of the particle at time t , we make use of the Fundamental Theorem of Calculus as follows:

$$\begin{aligned}
 s(t) &= s(0) + \int_0^t v(x) dx \\
 &= 6 + \int_0^t 2 + x - \sin(x) dx \\
 &= 6 + \left(2x + \frac{1}{2}x^2 + \cos(x) \right) \Big|_0^t \\
 &= 6 + \left[\left(2t + \frac{1}{2}t^2 + \cos(t) \right) - \left(2(0) + \frac{1}{2}(0)^2 + \cos(0) \right) \right] \\
 &= 6 + \left[\left(2t + \frac{1}{2}t^2 + \cos(t) \right) - \cos(0) \right] \\
 &= 6 + \left(2t + \frac{1}{2}t^2 + \cos(t) - 1 \right) \\
 &= 5 + 2t + \frac{1}{2}t^2 + \cos(t)
 \end{aligned}$$

So, the position of the particle at time t is $\left(5 + 2t + \frac{1}{2}t^2 + \cos(t) \right) ft$.

Problem 8**Part (a)**

Using the fact that the area of a right triangle with base of length b and height of length h is given by $A = \frac{1}{2}bh$, we see that

$$\int_0^2 f(x) dx = 1/2(2)(6) = 6.$$

Part (b)

Using the fact that the area of a half circle with radius r is given by $A = \frac{1}{2}\pi r^2$, we see that

$$\int_2^6 f(x) dx = \frac{1}{2}\pi(2)^2 = 2\pi.$$

Part (c)

To compute this integral, we will use the fact that the area of a right triangle with base of length b and height of length h is given by $A = \frac{1}{2}bh$, as well as the fact that the area of a rectangle with base of length b and height of length h is given by $A = bh$. Noting that this area is below the graph, we see that

$$\int_6^{10} f(x) dx = \int_6^8 f(x) dx + \int_8^{10} f(x) dx = -\left(\frac{1}{2}(2)(2) + (2)(2) \right) = -6.$$

Part (d)

Using the fact that $\int_0^{10} f(x) dx = \int_0^2 f(x) dx + \int_2^6 f(x) dx + \int_6^{10} f(x) dx$, we may see that

$$\int_0^{10} f(x) dx = 6 + 2\pi - 6 = 2\pi.$$

Problem 9

- (a) F
- (b) F
- (c) F
- (d) T
- (e) T
- (f) F

Problem 10

Use a resource like www.integral-calculator.com to check your answers for this section.